DE–2111

Sub. Code
11

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Write short note on :

   (a) Basic elements of a Computer and Communications System. (6)

   (b) Connectivity and Interactivity. (6)

   (c) Generations of Computers. (8)

2. (a) Explain the four types of applications software. (10)

   (b) Explain the Spreadsheets and Database software. (10)

3. Write short note on :

   (a) Video conferencing and picture phones (5)

   (b) Workgroup computing (5)

   (c) Analog and Digital Signals (5)

   (d) Local Networks. (5)

4. (a) Explain the Compression and Decompression. (10)

   (b) Explain the Hard Disks and Magnetic Tapes. (10)

5. (a) Explain the two types of Database Organization. (10)

   (b) Explain the Features of DBMS. (10)

6. (a) Discuss the six phases of system analysis and design. (10)

   (b) Explain the Five Generations of programming Languages. (10)

7. (a) Explain the Management Information Systems. (10)

   (b) Explain the HTML and XML. (10)

8. (a) Explain the Word processing and Presentation graphics software. (10)

   (b) Explain the Internet and Web browsers. (10)

DE–2112

Sub. Code
12

DISTANCE EDUCATION

P.G.D.C.A. DEGREE EXAMINATION, DECEMBER 2015.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Subtract 17 from 55 by using 1's complement method. (5)

   (b) Perform the following :

   (i) 1101 + 111101 in binary. (5)

   (ii) 556 + 724 in octal. (5)

   (iii) 87 – 49 in Hexadecimal. (5)

2. Simplify using Karnaugh map and implement using NAND gates only

   $Q(A, B, C, D) = \Sigma m\,(0, 2, 4, 7, 8, 11, 12, 14)$. (20)

3. (a) Draw the circuit of a half subtractor and give its truth table. (8)

   (b) Explain the parallel binary adder. (12)

4. (a) Draw the circuit of a shift register and explain its working. (10)

   (b) Explain the working of magnetic disk. (10)

5. (a) What is DMA? Explain. (10)

   (b) Explain the operation in a fetch cycle. (10)

6. (a) Draw a down counter circuit and explain its functions with truth table. (14)

   (b) Explain : Dynamic RAM. (6)

7. (a) What is an encoder? Explain. (10)

   (b) Implement using fundamental gates.

   $Q = A\,\overline{B}\,\overline{C} + \overline{AB}\,C + A\,\overline{B}\,C$. (10)

8. Discuss the following :

   (a) Character codes. (5)

   (b) Demultiplexer. (5)

   (c) Signed 2's complement representation. (5)

   (d) SoP notation. (5)

DE–2113

Sub. Code
13

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Academic year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Describe the Input/Output statements available in C. (12)

   (b) Explain and compare the break and continue statement. (8)

2. (a) Explain the syntax of function declaration and invocation in C language. (10)

   (b) How are dimensional array elements are read and written? (10)

3. (a) What are pointers? When pointers are useful? Why it is necessary to declare the pointer type variable? (12)

   (b) Explain the concept of structures with example. (8)

4. (a) Write a program to convert the given decimal number into an equivalent binary number. (10)

   (b) Write a program in C to perform matrix multiplication. (10)

5. (a) Write a recursive function to evaluate $n!$ and write a program to evaluate

$$nC_r = \frac{n!}{r!(n-r)!}.$$ (10)

(b) Two data files 'Ladies' and 'Gents' contain sorted list of names of girls and boys in a class respectively. Write a program to produce a third file 'students' which holds the merged list of two given files. (10)

6. (a) Classify the data structure and explain. (8)

(b) Give an algorithm for adding and deleting an element in a circular queue and explain. (12)

7. (a) Explain the structure of a singly linked list. (8)

(b) Discuss the application of trees. (12)

8. (a) Describe the operations that can be performed on stack with examples. (12)

(b) Explain the linked representation of a binary tree. (8)

DE–2114

Sub. Code
14

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the following concepts :

   (i) Working with windows elements (7)

   (ii) Windows Explorer (3)

   (b) Explain the Office Shortcut Bar (6)

   (c) Give any four applications of MS OFFICE. (4)

2. (a) How to create, modify, delete rows and columns of the table? Explain. (8)

   (b) How to change the text styles? Explain. (4)

   (c) Explain the bullet and number list options. (8)

3. (a) Write short notes on the Mail merge. (10)

   (b) How to create the mathematical formula in the MS Word? Explain. (6)

   (c) Explain the Headers and Footers features. (4)

4. (a) How to enter, modify, insert, delete and copy data cell in the excel sheet? Explain (10)

   (b) How to create the chart? Explain any two types of charts. (10)

5. (a) How to add, edit and delete records in the database? Explain. (8)

   (b) How to search and sort the database? Explain. (6)

   (c) How to create the database? Explain with an example. (6)

6. (a) Explain the following : (10)

   (i) Create and edit the power point presentation

   (ii) Add and delete a slide

   (iii) Change the views.

   (b) How to create graph charts and implemented in the power point presentation? Explain. (10)

7. (a) Give suitable operations sequence to do the following. (10)

   (i) Insert a picture in the slide

   (ii) Insert sound in the slide

   (iii) Animated the pictures.

   (b) Explain the various data types in MS access. (10)

8. (a) How to create, save and edit the forms? Explain. (10)

   (b) How to create and print the reports of Sales Bill? Explain. (10)

DE–2115

Sub. Code
21

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is Object Oriented Programming? What are its advantages?

   (b) What are manipulators? Demonstrate with an example, the need for 'setw' manipulator.

2. (a) Explain with examples, the looping structures in C++.

   (b) Explain with examples, the uses of enumerated data types.

3. (a) Write a program in C++ that counts the number of words and number of characters in a phrase typed in by the user.

   (b) Explain with examples, overloading unary operators and overloading binary operators.

4. (a) Write short notes on virtual functions.

   (b) Write a program in C++ to solve the quadratic equation $ax^2 + bx + c = 0$ and print the roots.

5. (a) What are the advantages of function prototypes in C++?

   (b) Explain in detail, the formatted console I/O operations in C++. Give examples.

6. Explain with examples, the different forms of inheritance.

7. (a) Write short notes on pointers to objects.

   (b) Explain how a portion of an array can be passed onto a function.

8. (a) Explain the types of constructors with examples.

   (b) What is a file? Write a program in C++ to process an employee file.

DE–2116

Sub. Code
22

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the System Concepts. (10)

   (b) Explain the any two types of systems (10)

2. (a) Explain the MIS and TPS. (10)

   (b) Explain the interpersonal skills and communication skills. (10)

3. (a) How to determining the users information requirements? Explain. (10)

   (b) Explain any two fact finding techniques. (10)

4. (a) What are steps in feasibility analysis? Explain. (10)

   (b) What is meant by data flow diagram? Explain. (10)

5. (a) Explain the input design. (10)

   (b) Explain the Classification of forms. (10)

6. (a) Explain the types of files. (10)

   (b) Explain the types of databases. (10)

7. (a) Explain the top down approach. (10)

   (b) Explain the system testing. (10)

8. (a) Discuss the system maintenance. (10)

   (b) Discuss the procedure of hardware and software selection. (10)

DE–2117

Sub. Code
23

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Give the purpose of Tool Box. (10)

   (b) What is meant by Visual Basic Environment? (10)

2. (a) How to start with Visual Basic? (10)

   (b) Discuss the technique of printing a form. (10)

3. (a) How to create controls in Tool Box? (10)

   (b) Give the purpose of

      (i) Text Box

      (ii) Label Box (10)

4. (a) Discuss the operators used in Visual Basic. Explain its execution with examples. (10)

   (b) Explain the execution of If… Then… Else statement. (10)

5. (a) Discuss the following :

   (i) Error Trapping

   (ii) Object Browser. (10)

   (b) What is Animation? How to handle it in Visual Basic? (10)

6. (a) Explain the technique of creating classes and menus. (10)

   (b) Discuss the importance of Debugging Tools. (10)

7. (a) Explain the following :

   (i) Date Functions

   (ii) Financial Functions. (10)

   (b) How static arrays and dynamic arrays are defined and used? Explain them with examples. (10)

8. (a) How sequential files differ from Random Access Files and Binary Files? (10)

   (b) Explain the technique of building ActiveX controls. (10)

DE-2118

Sub. Code
24

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. With neat diagrams, explain the working principle of a cathode Ray tube used for displaying objects. (20)

2. With algorithm, explain the procedure of generating a circle. (20)

3. Explain the procedure of manipulating objects with concatenation of transformations. Also illustrate with matrix representation. (20)

4. (a) Write about Window–to–viewport coordinate transformation. (8)

   (b) Discuss the procedure of Sutherland – Hodgeman polygon clipping technique. (12)

5. Give a detailed description on general three – dimensional rotations transformation. Explain it with neat diagram and matrix representations. (20)

6. Summarize the procedure of two projecting methods applied to three dimensional objects. (20)

7. Write notes on :

   (a) Depth – Buffer algorithm. (10)

   (b) Depth – sorting method. (10)

8. Describe the components of user interface design and the command languages used on it. (20)

DE–9014

Sub. Code
15

DISTANCE EDUCATION

P.G.D.C.A.(S) EXAMINATION, DECEMBER 2015.

First Semester

LAB – I : DATA STRUCTURES USING 'C'

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by Lot system.

Each subdivision carries 50 marks.

1. (a) With a C program to find the roots of a quadratic equation with all possibilities.

$ax^2 + bx + c = 0$.

(b) Write a C program to implement PUSH and POP operations on stack.

---------- Cut here ----------

2. (a) Write a C program for matrix manipulation (Addition, subtraction and multiplication) using function.

(b) Write a menu driven program in C to find an element using linear and binary search.

3. (a) Three are 20 students in a class. Their names and marks in five different subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS' . Otherwise 'FAIL'. Write a C program to do the above using structure concept.

(b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

4. (a) Write a C program to find the roots of a quadratic equation $ax^2 + bx + c = 0$ using function definition.

(b) Write a C program to implement insert and delete operations on linked list.

---------- Cut here ----------

5. (a) Write a C program to implement all string operations using switch statement.

(b) Write a C program to implement insert and delete operations on queue using array concept.

---------- Cut here ----------

6. (a) Write a C program to copy the contents of one file to another file.

(b) Write a C program to evaluate the given Mathematical expression using stack.

7. (a) Write a C program to check whether the given word is Palindrome or note.

   (b) Write a menu driven program in C to sort the given set of elements using insertion and selection sort.

---------- Cut here ----------

8. (a) Write a C program to sort 10 numbers in ascending order with naming of variable and the value before and after sorting.

   (b) Write a C program to implement insert and delete operations on stack.

DE–9015

Sub. Code
16

DISTANCE EDUCATION

P.G.D.C.A. EXAMINATION, DECEMBER 2015.

First Semester

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 academic year)

Time : Three hours Maximum : 100 marks

Each question should be selected by the Examiner and given to the candidates by lot system.

Break-Up of Mark	:	
Record Note Book	:	5
Algorithm/Flow chart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Prepare the first page of M.C.A. PRATICAL RECORD NOTE book with picture insertion and alignment.

   (b) Create the following worksheet in EXCEL for Electricity Bill

CONSUMER NO.	TYPE	NAME	BILL	PMR	CMR	NO. UNITS
101	D	RAM	545	645	100	

(i) Add data for 10 customers with type 'D' for domestic user and type 'I" for Industrial user.

(ii) All the consumers should be charged a minimum bill of Rs. 20 even if their bill amount is below Rs. 20

(iii) Find UNITS column for each customer and calculate bill using the slab given below :

TYPE D CONSUMER

UNITS CONSUMED	RATE/UNIT Rs.
FIRST 100	0.80
NEXT 200	1.25
REMAINING	2.00

TYPE I CONSUMER

UNITS CONSUMED	RATE/UNIT Rs.
FIRST 100	1.10
NEXT 200	2.40
REMAINING	3.50

---------- Cut here ----------

2. (a) Create a worksheet in MS-Excel with following columns :

Employee number, Employee Name, designation, Basic pay, HRA, DA, LIC, PF, Gross pay, Net pay

(i) Type data for empno. Empname, designation, Basic pay and LIC, PF.

(ii) Calculate HRA = 20% of Basic

DA = 30% of Basic

Gross pay Basic + HRA+ DA

Net pay = Gross pay – (LIC+PF)

(iii) Draw the bar chart between emp name and Netpay

(iv) Sort the designation column and employee column name at a time

(b) Create HTML page to display the names of any three eminent sports person. Link to another page when the user clicks over the name. The linked page should display features of the person clicked.

---------- Cut here ----------

3. (a) Create the following Inventory worksheet in MS-EXCEL :

ITEMS NO.	NAME	PRICE	QUANTITY
101	Bolts	200	1000
STOCK	REORDER	ISSUED	QUANTITY ON HAND
500	800	200	

(i) Enter all the data items except QUANTITY ON HAND for 10 items.

(ii) Find QUANTITY ON HAND using the formula

STOCK QUANTITY = QUANTITY ON HAND + QUANTITY PURCHASED – QUANTITY ISSUED

Find tool stock value in inventory as a product of total quantity hand and total price (Σ quantity on hand * Σ price). Display all the items in red color whose quantity on hand is below reorder level.

(b) Create HTML page to display 10 courses and their highlighted features, conducted by an educational institution in a colorful format.

4. (a) Create two worksheets containing day to day house hold expenses for the months January and February 2005, with the following columns :

ITEM	AMOUNT
Stationery	25.00

(i) Add 10 different items as shown above, for two months in two different worksheets

(ii) Consolidate both the months data and find the total expenses on each items

(iii) Find the total expenses for two months

(iv) Find the maximum and minimum expense amount.

(b) Create STUDENT table with the following fields REGNO, NAME, MARK1, MARK2, MARK3. Create ADDRESS table with fields REGNO, STREET, CITY and PIN.

(i) Write a query to display REGNO, NAME and total of all the three subject marks

(ii) Write a query to display REGNO, NAME, STREETM, CITY PIN and total of all the three subject marks.

---------- Cut here ----------

5. (a) Create two worksheets containing day to day house hold expenses for the months January and February 2005, with the following columns :

ITEM	AMOUNT
Stationery	25.00

(i) Add ten different items as shown above, for two months in two different worksheets

(ii) Consolidate both the months data and find the total expenses on each items

(iii) Find the Total expenses for two months

(iv) Find the maximum and minimum expense amount.

(b) Create HTML page to display any two pictures. Link to another page when user clicks over the picture. The linked page should display features of the picture clicked.

6. (a) Draw Line, Bar, PIE charts for the data given below :

ABC COMPANY LIMITED

YEAR	SALES	COST	PROFIT
2001	1000	400	600

Add data for ten years from 2001 to 2010. Provide titles, legends, grids and data labels.

(b) Create HTML page that receives employee name, age, street, city, pin, data. Receive male/female and married/unmarried details through option buttons. Receive hobbies through check boxes.

---------- Cut here ----------

7. (a) ***Instruction : Open a New Database and add tables***

Create employee table with the fields EMPNO, NAME, AGE, SEX, STREET, CITY, PIN, SALARY

(i) Add data for 20 employees

(ii) Write a query to display all the male employees whose salary is between 1000 and 5000 and living in city "CHENNAI"

(iii) Write a query to display all the female employee whose ages are in the range 50-60

(iv) Show all the records in the table for the city "TRICHY" by filtering.

(b) Create HTML page that plays video and audio files.

8. (a) To create the document and type the following mathematical expression :

$$\int_{0}^{\infty} xdy/dx \sum_{i=1}^{100} ax_i + b + |A| \pm \sqrt[3]{ax + b\lambda}\ .$$

(b) Create a worksheet with the following columns

Salesman number, Salesman Name, City, Product Name, Sale Amount.

Add three records for five different salesmen who have carried out sales of different products in different cities. Find the following :

(i) City wise total sales

(ii) Sales man wise total sales

(iii) Production wise total sales

(iv) Grand total sales.

DE–9016

Sub. Code
25

DISTANCE EDUCATION

P.G.D.C.A (S) EXAMINATION, DECEMBER 2015.

Second Semester

COMPUTER LAB III — GRAPHICS USING C++

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

Break-up of mark :

Record Note Book : 5
Algorithm: 15
Program: 35
Debugging & Execution : 35
Result : 10
Total : 100

ONE questions should be selected by the examiner and given to the candidates by lot system.

1. Develop a C++ program to draw an isosceles triangle and rotate by 45°.

---------- Cut here ----------

2. Narrate the bresenham's line drawing algorithm to draw a straight line with a C++ program.

---------- Cut here ----------

3. Develop a C++ program to draw a non-linear object "Ellipse".

4. Write down the C++ program to draw a straight line using simple DDA technique.

---------- Cut here ----------

5. Write down the C++ program to draw a circle, divide it into four quadrant and file with different colors to distinguish the quadrants.

---------- Cut here ----------

6. Develop a C++ program to implement the Bresenham's algorithm for drawing circle and ellipse.

---------- Cut here ----------

7. Write a C++ program to draw an equi-lateral triangle that coincides with the side of a square (i.e) base to the tri-angle = side of the square. Rotate it by 180°.

---------- Cut here ----------

8. Develop a C++ program to draw a ball perform the linear movement and wave movement.

---------- Cut here ----------

9. Write a C++ program to draw a circle whose diameter is equal to side of a square.

---------- Cut here ----------

10. Develop a C++ program to implement the polygon clipping algorithm using graphics mode functions.

DE–9017

Sub. Code
26

DISTANCE EDUCATION

P.G.D.C.A. (S) DEGREE EXAMINATION, DECEMBER 2015.

Second Semester

COMPUTER LAB IV — VISUAL BASIC

(2005 to 2010 Academic Year)

Time : Three hours Maximum : 100 marks

One question should be selected by the examiner and given to the candidate by Lot system.

Break-Up of Marks

Record Note Book	: 5
Algorithm/Flow chart/ Syntax	: 15
Writing Program/ Coding	: 35
Debugging and Execution	: 30
Result/output	: 10
Viva–voce	: 5
Total	: 100

1. Write a VB project that receives a year number from a text box and month name from list box and displays number of days in the given month. Take care of leap years. Use Lost Focus Event for list box.

---------- Cut here ----------

2. Write a VB project that stores 10 employee records with fields EMPNO, NAME, AGE, SEX, SALARY in an array. Display data fields in text boxes and provide command buttons to move to desired record.

3. Write a VB project that receives a foreign currency value selected from a list box and converts it into equivalent Indian rupees.

---------- Cut here ----------

4. Write a VB project using control array that creates a scientific calculator with appropriate command buttons.

---------- Cut here ----------

5. Write a VB project to create a screen saver that displays a list of pictures with one second pause in between successive pictures.

---------- Cut here ----------

6. Write a VB project for commercial bank operations using SB account database with the following features.

   (a) ADD NEW ACCOUNT

   (b) DEPOSIT AMOUNT

   (c) WITHDRAW AMOUNT

   (d) Calculate simple interest and update balance taking average of last 6 month balance in the account.

   (e) Close account.

---------- Cut here ----------

7. Write a VB project using built in AX control (Rich Text Box), develop the windows NOTEPAD like editor with file and edit menus and also display the floating menu, whenever necessary.

8. Write a VB project for a blood bank that maintains a list of donors with addresses and their blood group. Provide the following reporting features

   (a) Search and display the address of a particular donor

   (b) Display all the donors.

      (i) in age group 20–30

      (ii) in particular city

      (iii) with particular blood group.

---------- Cut here ----------

9. Write a VB project using AX DLL or EXE add a class module that would perform that following functions.

   (a) finding the factorial of the given number

   (b) sum of digits.

---------- Cut here ----------

10. Write a VB project using Active X control to create a Text Box that accepts only numeric value

    Provide the following properties for the Text Box :

    Back color, fore color and text.